

Conditional Credit Pricing as Bayesian Default Updating: The Correlated-Signal Problem

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Abstract

A credit spread encodes a prior probability of default; news about an issuer is evidence; the fair default probability is the Bayesian posterior. We make this correspondence explicit and operational, mapping spreads to priors through the credit triangle and revising them with likelihood ratios estimated from history. We then isolate the failure mode that a naïve implementation invites: because credit signals such as a downgrade and an earnings miss are typically driven by a common deterioration, multiplying their likelihood ratios as if they were independent overstates default risk. On a synthetic issuer panel the product of marginal likelihood ratios is 8.5 where the joint ratio is only 3.5, and a name carrying all three signals receives a naïve posterior of 14.8% against a correlation-aware 6.7%. A single population-wide joint ratio is not a remedy either: with credit rating as a confounder it understates. A hierarchical Bayesian logistic model that conditions on rating and treats the signals jointly is the best calibrated out of sample (Brier 0.0575 versus 0.0591 naïve and 0.0657 prior-only). The exercise reframes a standard probability puzzle as the daily mechanics of a credit relative-value desk.

1 Introduction

The classic medical-test puzzle—a positive result on a rare condition leaves the patient probably healthy—and the pricing of an issuer’s default risk are the same object [Gigerenzer and Hoffrage, 1995]. Both begin with a base rate, observe evidence, and update. In credit the base rate is not invented: it is quoted by the market through the spread. Under a reduced-form view [Jarrow and Turnbull, 1995, Duffie and Singleton, 1999] the spread implies a default intensity, hence a prior default probability. Each subsequent piece of news—a rating downgrade, an earnings miss, a spread move—is evidence with a likelihood ratio, and Bayes’ rule in odds form prescribes exactly how the implied default probability should move. If the market price has not moved as much as the likelihood ratios imply, there may be a trade.

This paper sets out that correspondence and then studies the complication that makes it nontrivial in practice. Real credit signals are not conditionally independent: the same underlying deterioration tends to trigger several of them at once. Treating them as independent confirmations—multiplying likelihood ratios—double-counts the shared information and produces overconfident default probabilities. We quantify the effect and show that the natural fix must account both for the correlation among signals and for rating heterogeneity, which a hierarchical Bayesian model does.

2 The framework

Prior from the spread. For a constant default intensity λ and recovery R , the credit triangle gives $\lambda = s/(1 - R)$ for a spread s , a survival probability $S(T) = e^{-\lambda T}$, and a prior default

probability

$$\pi = \mathbb{P}(\text{default}) = 1 - e^{-\lambda T} = 1 - \exp\left(-\frac{s}{1-R} T\right). \quad (1)$$

For $R = 0.4$ and $T = 1$, a 121 bp spread implies $\pi \approx 2.0\%$.

Evidence and likelihood ratios. For a binary signal E and default indicator D , the likelihood ratio is $\text{LR}(E) = \mathbb{P}(E | D) / \mathbb{P}(E | \neg D)$, estimated from historical frequencies with Laplace smoothing.

Updating in odds form. With prior odds $\pi / (1 - \pi)$ and conditionally independent signals $E_1, \dots, E_k$,

$$\underbrace{\frac{\mathbb{P}(D | E_{1:k})}{\mathbb{P}(\neg D | E_{1:k})}}_{\text{posterior odds}} = \underbrace{\frac{\pi}{1 - \pi}}_{\text{prior odds}} \prod_{j=1}^k \text{LR}(E_j), \quad \mathbb{P}(D | E_{1:k}) = \frac{\text{odds}}{1 + \text{odds}}. \quad (2)$$

Equation (2) is the engine: prior from the spread, multiplied by the likelihood ratio of each new signal.

3 The independence trap

The product in (2) is exact only when the signals are conditionally independent given D . Credit signals violate this. Writing the correlation-aware *joint* likelihood ratio $\text{LR}(E_1 \cap \dots \cap E_k) = \mathbb{P}(E_1 \cap \dots \cap E_k | D) / \mathbb{P}(E_1 \cap \dots \cap E_k | \neg D)$, positive dependence makes $\prod_j \text{LR}(E_j) > \text{LR}(E_1 \cap \dots \cap E_k)$, so the naïve update overstates the posterior. Table 1 shows the gap on the synthetic panel of Section 5: three signals with marginal ratios near 2 multiply to 8.5, while their joint ratio is only 3.5. Starting from a 2% prior, a name carrying all three is assigned 14.8% naïvely but 6.7% once the redundancy is removed.

Quantity	Value
LR(downgrade)	2.12
LR(earnings miss)	1.96
LR(spread widening)	2.04
Product of marginals	8.50
Joint LR (all three)	3.54
Naïve posterior PD (2% prior)	14.8%
Correlation-aware posterior PD	6.7%

Table 1: Marginal likelihood ratios multiply far above the joint ratio, so the independence assumption inflates the posterior.

4 A hierarchical Bayesian model

A single population-wide joint ratio is not a sufficient remedy, because credit rating confounds the relationship: pooling across ratings makes the joint ratio understate risk for any given rating. We therefore model the default outcome of issuer i with a hierarchical logistic specification,

$$\text{logit } p_i = a_{r(i)} + \sum_k \beta_k x_{ik}, \quad a_r \sim \mathcal{N}(\mu_a, \sigma_a), \quad \beta_k \sim \mathcal{N}(0, 1.5), \quad D_i \sim \text{Bernoulli}(p_i), \quad (3)$$

with partial pooling of the rating intercepts a_r and signals x_{ik} entered jointly, so each $\exp(\beta_k)$ is a rating- and co-signal-adjusted odds multiplier. Inference is by NUTS in PyMC [Abril-Pla et al., 2023, Gelman et al., 2013].

5 Synthetic experiments

Each issuer draws a rating, a latent deterioration $z \sim \mathcal{N}(0, 1)$, and a default outcome with logit $p = a_r + \gamma z$. The three signals load on the same z , inducing the cross-signal correlation that defeats independence, while the observed spread is generated from the rating-implied prior plus noise (a deliberately stale prior, so the signals carry new information). We fit on 60% of 6,000 issuers and evaluate on the rest.

6 Results

Table 2 reports holdout performance for three predictors: the spread prior alone, the naïve independent update, and the Bayesian posterior. Updating on signals helps (AUC $0.767 \rightarrow 0.837$); accounting for their correlation and for rating heterogeneity helps further and improves calibration (Brier $0.0591 \rightarrow 0.0575$, log loss $0.2100 \rightarrow 0.2056$). Figure 1 shows the naïve curve bowing above the diagonal in the high-probability region—overstated risk—while the Bayesian curve tracks the diagonal. On names carrying two or more signals the naïve predictor runs hotter than the correlation-aware model (17.4% versus 13.9%), the redundancy bias made concrete.

Predictor	AUC	Brier	Log loss
Prior-only (spread)	0.767	0.0657	0.2458
Naïve independent LR	0.837	0.0591	0.2100
Hierarchical Bayesian	0.842	0.0575	0.2056

Table 2: Out-of-sample performance ($n = 2,400$ holdout issuers).

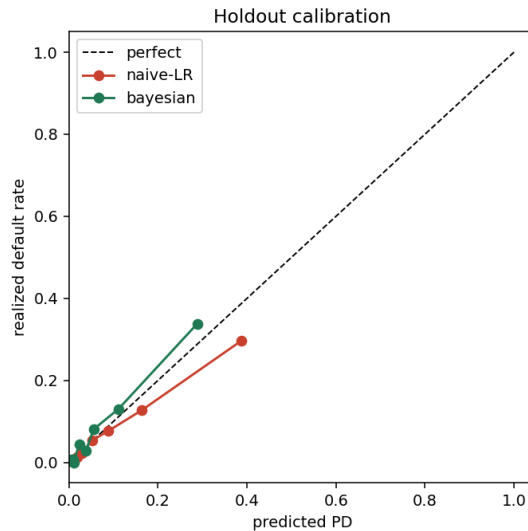


Figure 1: Holdout calibration. The naïve update (red) overstates default probability for high-risk names; the hierarchical Bayesian model (green) is closer to the diagonal.

7 Discussion

The contribution is conceptual as much as empirical: the spread-implied prior times likelihood-ratio update is the mechanics of conditional credit pricing, and it is the same operation as the textbook test-positive puzzle. The practical caution is that signals are correlated, so the independence shortcut produces overconfident default probabilities; the appropriate correction conditions jointly on the signals and on rating. Limitations are clear: the data are synthetic, the signals binary and static, and the model omits the term structure of hazards. Natural extensions are real rating-migration and CDS data, continuous and time-stamped signals, a copula treatment of the joint likelihood, and a mapping from posterior default probabilities back to fair spreads to define explicit relative-value signals.

References

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